Data Engineering Applied Field Engineering Role Panel Interview Scenario

Your Role: Data Engineering Applied Field Engineer (AFE)

You are representing Snowflake as an Applied Field Engineer. Your objective is to partner with the Tasty Bytes team (customer) to design a solution that solves their immediate technical debt while demonstrating why Snowflake is the superior platform for their global expansion. You must move beyond "how it works" to "why it matters" for their business.

Objective

Propose and demonstrate a scalable Data Engineering strategy for Tasty Bytes. This solution must enable their growth to 1,120 trucks and $320 million in sales while transitioning from on-premises infrastructure to a modern cloud platform (AWS, Azure, or GCP). *Note: While Snowflake is the primary focus of this AFE role, the solution should involve technologies you know and understand today. We are evaluating how you architect a solution, present your logic, and tell a convincing story to win over the customer.*

Current IT Landscape

- **Data Silos:** Information is scattered across various applications and formats. Sources include PostgreSQL (sales transactions) and MySQL (product inventory).
- **High Latency:** Data is currently loaded only once per day, leaving the business with no insight into real-time operations.
- **Wasted Data Assets:** Website clickstream data is collected but remains entirely unused in the current architecture.
- **Reliability Gaps:** The legacy on-prem IT stack is frequently unreliable, causing downtime that results in lost revenue.
- **Privacy Concerns:** There are existing complaints that customer privacy has not been sufficiently protected in the legacy system.
- **Cloud Migration:** Management requires a move from on-premises infrastructure to a major cloud provider (AWS, Azure, or GCP).
- **Open Standards:** The customer prefers all data to be stored in interoperable open data formats to avoid vendor lock-in.

Key Requirements

- **Unified Platform:** Integrate all data sources into a single, unified cloud data platform.
- **Open Lakehouse:** Implement a lakehouse architecture using open table formats such as Apache Iceberg.
- **Low Latency:** Target an ingestion latency of 1 minute or less on raw data.
- **Automation:** All pipelines must be automated via reliable orchestration with full version control and CI/CD practices.

Requirement 1: Architecture Strategy (The Pitch)

- **The Winning Story:** Sell the panel on why Snowflake is the right choice for Tasty Bytes. Focus on ROI, reduced complexity, and speed to market.
- **Unified Integration:** Propose a plan to ingest PostgreSQL, MySQL, and Clickstream data seamlessly into the Snowflake Data Cloud.
- **Open Lakehouse (Iceberg):** Explain your implementation of Apache Iceberg tables to ensure data interoperability and eliminate vendor lock-in fears.
- **Low Latency Strategy:** Detail the strategy for achieving the 1-minute latency target using Snowpipe Streaming v2.
- **Simplified Connectivity:** Discuss the value of using Snowflake Openflow to manage complex ingestion without managing underlying infrastructure.
- **Security & Scale:** Explain how the solution addresses global scale and customer privacy via Snowflake Horizon.

Requirement 2: Technical Deep Dive (The Demo)

- **Streaming Pipeline:** Demonstrate a data pipeline ingesting high-volume logs or transactions into Snowflake Iceberg tables. *Note on Implementation: To show low-latency ingestion, we recommend using streaming tools. However, if you lack access to external cloud providers for the demo, please use what you have available to you to best simulate the scenario.*
- **Advanced Transformation:** Showcase the use of Dynamic Tables or Snowpark to transform raw data into a harmonized analytical layer.
- **Orchestration & DevOps:** Show how the entire pipeline is automated via reliable orchestration (e.g., Snowflake Tasks) with full version control.
- **Data Quality:** Demonstrate integrated data quality checks that ensure the platform provides accurate insights for supply chain and sustainability reporting.

Suggested Presentation Format (90 Minutes)

- **05 minutes:** Introduction and Context.
- **05 minutes:** Acknowledge Business Goals. Confirm specific data engineering needs.
- **20 minutes:** Requirement 1: Architecture Pitch. Present your conceptual design and tell a winning story as a Snowflake AFE.
- **40 minutes:** Requirement 2: Technical Demo. Show the live ingestion, transformation, and automation code demo.
- **10 minutes:** Implementation Roadmap. Outline the path to production, governance, and implementation best practices.
- **10 minutes:** Questions, Answers, and Closing. Final summary of the Snowflake value proposition.

Evaluation Criteria

- **Strategic Leadership:** Ability to choose the right Snowflake features and justify architectural decisions based on business value.
- **Technical Breadth:** Hands-on skills in ingestion, transformation, and governance using modern features like Iceberg and Snowpipe Streaming.
- **Stakeholder Engagement:** Ability to address the needs of different roles within the organization and handle technical objections or "curveball" questions with a consultative approach.
- **Presentation and Communication Skills:** Ability to command the room, use storytelling to bridge technical gaps, and pivot your language to address different stakeholders
- **Mission Alignment:** How well the technical solution supports the vision of becoming a sustainable, profitable global food truck network.

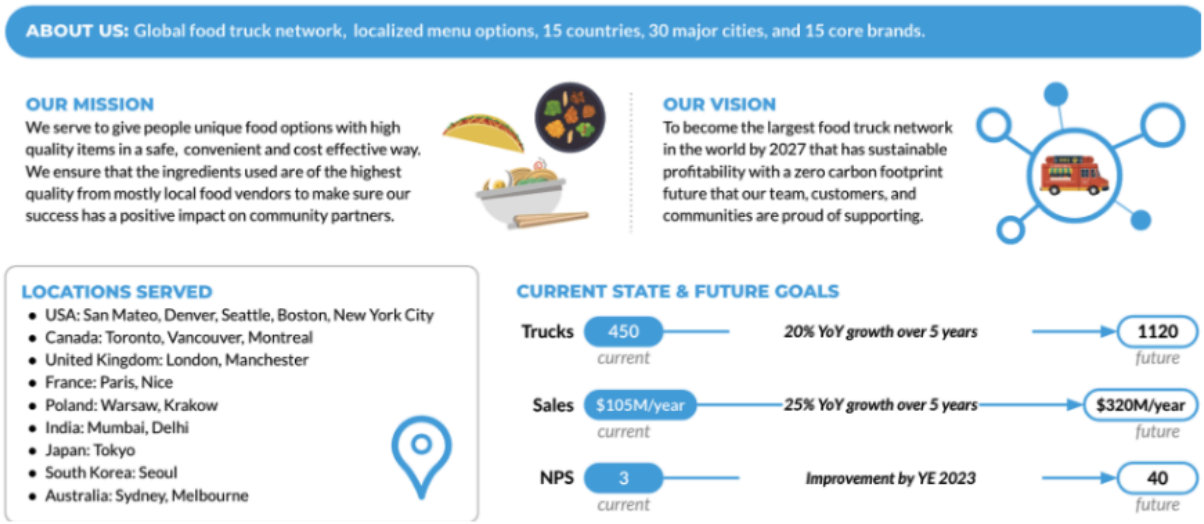
Guidelines

It’s up to you on how best to build a presentation flow to highlight your core strengths as a Applied Field Engineer.

Mock Customer: Tasty Bytes

You can find many details on who Tasty Bytes is as well as access to data and existing use-cases on the Snowflake Quickstarts website here.

Who is Tasty Bytes?



Starting point

- **Be aware of the audience**

Assign roles to panel members to mimic the Tasty Bytes team (e.g., CEO, CTO, Data Analyst, Supply Chain Manager, Sustainability Officer). Tailor your messaging to cater to both technical and business audiences.

- **Use free trial accounts**

Please use the free trial accounts for Snowflake (link). Demonstrating Snowflake is not mandatory but a technical demo is.

- **Get creative**

This is your chance to stand out and display the breadth and depth of your knowledge. It's up to you to stick to a bare minimum solution or go all out with the bells and whistles by including other products, services, and frameworks.

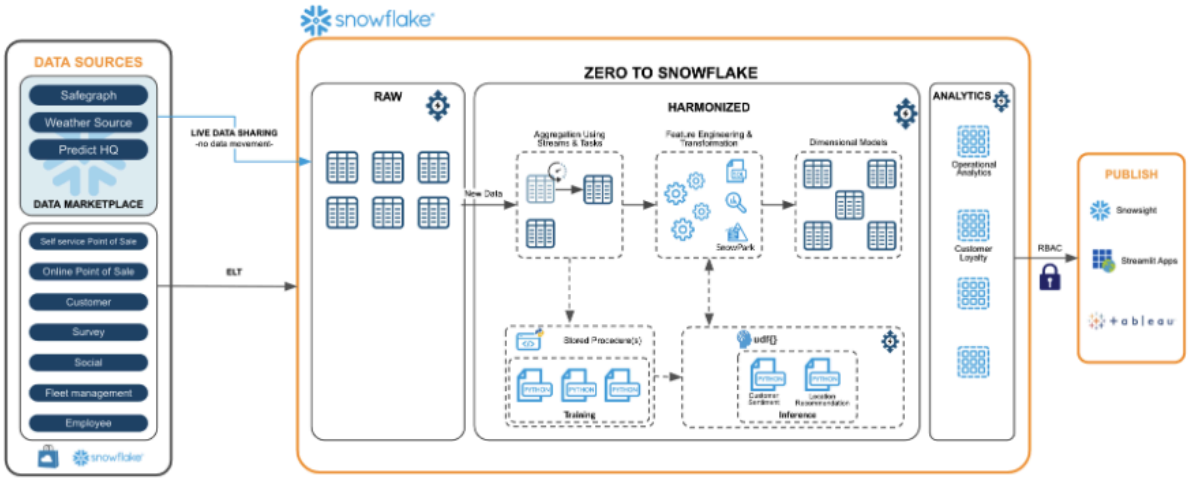
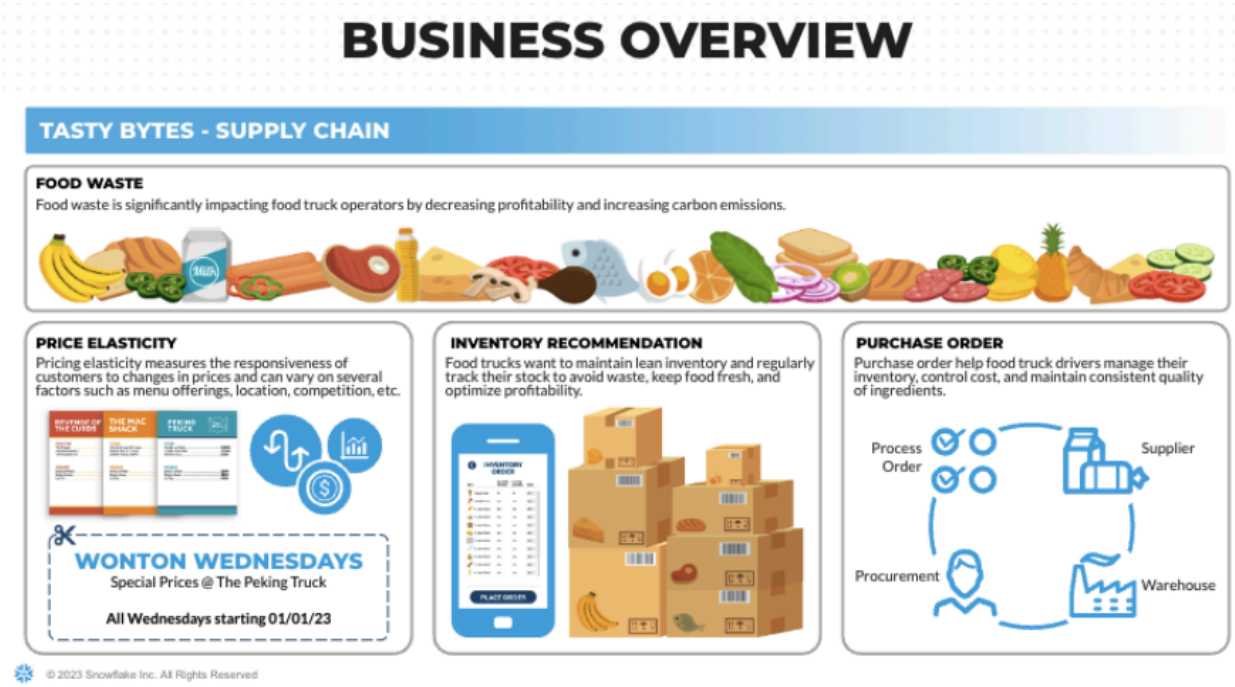
- **Tasty Bytes**

Feel free to add onto/ modify the Tasty Bytes Mission/ Vision, just be sure to set the stage before your presentation.

***Note Tasty Bytes is a reference point. Feel free to use any solution that you have built in the past to articulate your strengths, overall however please follow the agenda outlined in this exercise.*

Appendix

Example technical architecture



Example business overview